

A COMPARISON THEOREM FOR THE HOFSTADTER–CONWAY \$10,000 SEQUENCE AND ITS ALTERNATING VARIANT

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ABSTRACT. Let c be the Hofstadter–Conway \$10,000 sequence, $c(n) = c(c(n-1)) + c(n - c(n-1))$, and let s be the sequence defined by $s(n) = n - s(s(n-1)) - s(n - s(n-1))$, both with initial values 1, 1 (OEIS A004001 and A287422). Alkan conjectured in 2020 that $|s(n) - n/2| \leq c(n) - n/2$ for every $n \geq 1$. We prove the conjecture. On each dyadic block $[2^k, 2^{k+1}]$ both sequences are encoded by Dyck words, and the words for the next block are produced by two explicit threshold operators. The proof rests on a two-step domination lemma for symmetric Dyck words, which compares the operators after a small perturbation at the centre of the word.

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1. INTRODUCTION

The Hofstadter–Conway sequence

$$c(1) = c(2) = 1, \quad c(n) = c(c(n-1)) + c(n - c(n-1)) \quad (n \geq 3)$$

is sequence A004001 of the OEIS [5]. Conway stated that $c(n)/n \rightarrow \frac{1}{2}$ and offered a prize for a quantitative form of this statement, which was claimed by Mallows [3]. Kubo and Vakil [2] gave a detailed analysis of its structure on the dyadic blocks $[2^k, 2^{k+1}]$, on each of which $c(n) - \frac{n}{2}$ traces a positive arch.

Alkan [6] introduced the companion sequence

$$s(1) = s(2) = 1, \quad s(n) = n - s(s(n-1)) - s(n - s(n-1)) \quad (n \geq 3),$$

studied with related recursions in [1]. Its graph of $s(n) - \frac{n}{2}$ also consists of arches over the dyadic blocks, but their signs alternate. In a MathOverflow question [4] he conjectured, after checking all $n \leq 2^{32}$, that the arches of s never rise above those of c .

Theorem 1. *For every integer $n \geq 1$,*

$$\left| s(n) - \frac{n}{2} \right| \leq c(n) - \frac{n}{2}.$$

Equivalently, $n - c(n) \leq s(n) \leq c(n)$ for all n .

Outline. Encode the increments of c on the block $[2^k, 2^{k+1}]$ as a binary word w_k of length 2^k , and do the same for s , complementing the word on blocks where s lies below $n/2$; call the result v_k . Section 2 defines two operators F and G that double the length of a Dyck word, and Section 4 shows that $w_{k+1} = F(w_k)$ while v_{k+1} is $F(v_k)$ or $G(v_k)$ according to the parity of k . Theorem 1 becomes the statement that the lattice path of v_k never rises above that of w_k . Plain induction fails because G can overtake F in one step. The remedy, in Section 3, is to carry a stronger hypothesis through two steps at once: v_k lies below a perturbation $T(w_k)$ of w_k that lowers its path by one level away from the ends. The key lemma (Lemma 3.1) holds for Dyck words that are symmetric under reversal and complement, and all the words w_k have this symmetry. Figure 1 shows the three paths for $k = 5$.

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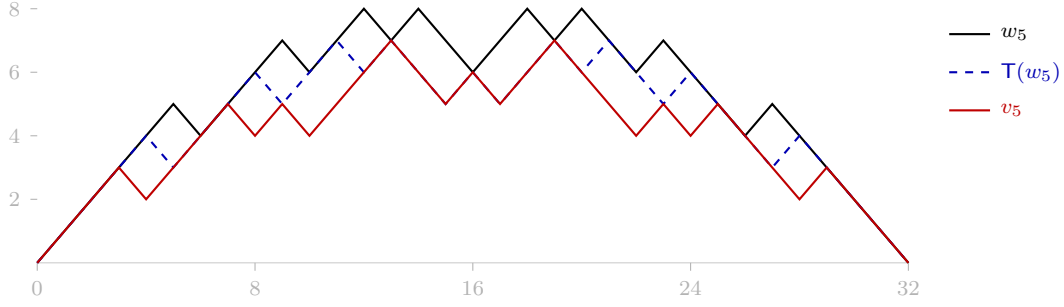


FIGURE 1. Heights of the block words for $k = 5$, that is, $2c(n) - n$ (black) and $|2s(n) - n|$ (red) for $32 \leq n \leq 64$, plotted against $n - 32$. The dashed path is $T(w_5)$, which lies between them, as the induction requires.

As by-products, the construction shows that s is well defined and *slow* (its increments are 0 or 1), that $s(2^k) = 2^{k-1}$, and that $s(n) - \frac{n}{2}$ is non-negative on $[2^k, 2^{k+1}]$ for even k and non-positive for odd k .

2. DYCK WORDS AND TWO THRESHOLD OPERATORS

2.1. Words. A binary word $u = u_1 \cdots u_L$ has prefix counts $P_u(i) = u_1 + \cdots + u_i$, with $P_u(0) = 0$. We write P when u is clear. Throughout, L is even and $m = L/2$. The word is a *Dyck word* if $2P(i) \geq i$ for all i and $P(L) = m$; its *height* at i is $2P(i) - i$. It is *irreducible* if $2P(i) > i$ for $0 < i < L$, and *symmetric* if $u_{L+1-i} = 1 - u_i$ for all i . For words of equal length we write $u \leq v$ if $P_u(i) \leq P_v(i)$ for every i . For a Dyck word, $u_1 = 1$ and $u_L = 0$.

Lemma 2.1. *A word u of length $L = 2m$ with $P(L) = m$ is symmetric if and only if*

$$(2.1) \quad P(L - i) = m - i + P(i) \quad (0 \leq i \leq L).$$

Proof. If u is symmetric, then $P(L) - P(L - i) = \sum_{s=1}^i u_{L+1-s} = \sum_{s=1}^i (1 - u_s) = i - P(i)$. Conversely, taking differences of (2.1) at $i - 1$ and i gives $u_{L+1-i} = 1 - u_i$. $\square$

2.2. The operators. Let u be a Dyck word of length L . For $0 \leq j \leq 2L$ let $I_j = [\max(0, j - L), \min(j, L)] \cap \mathbb{Z}$ and define

$$d_j(a) = a - P(a) - P(j - a) \quad (a \in I_j).$$

Set

$$A(j) = \min\{a \in I_j : d_j(a) \geq 0\}, \quad B(j) = \max\{a \in I_j : d_j(a) \leq 0\}.$$

Lemma 2.2 shows that A and B are the prefix counts of Dyck words of length $2L$, which we call $F(u)$ and $G(u)$.

The definition comes from the recurrences. When u encodes c on one dyadic block, the two earlier values that the recurrence for c reads on the next block sit at positions $a = A(j - 1)$ and $j - a$ of u , and (2.2) below is the recurrence itself (Proposition 4.1); $d_j(a)$ measures how far a candidate a is from that consistency. For example, take $u = 1100$, so $L = 4$ and $P = 0, 1, 2, 2, 2$. For $j = 3$ the values $d_3(a)$, $a = 0, 1, 2, 3$, are $-2, -2, -1, 1$, so $A(3) = 3$ and $B(3) = 2$; for $j = 4$ they are $-2, -2, -2, 0, 2$, so $A(4) = B(4) = 3$. Running over $j = 0, \dots, 8$ gives

$$A = 0, 1, 2, 3, 3, 4, 4, 4, 4, \quad B = 0, 1, 2, 2, 3, 3, 4, 4, 4,$$

so $F(1100) = 11101000$ and $G(1100) = 11010100$. These are the block words w_3 and v_3 of c and s on $[8, 16]$ (Section 4).

Lemma 2.2. *Let u be a Dyck word of length L .*

- (1) *Both thresholds exist; $d_j(A(j)) \in \{0, 1\}$ and $d_j(B(j)) \in \{-1, 0\}$.*
- (2) *$A(j), B(j) \geq j/2$, and $A(2L) = B(2L) = L$.*

(3) For $1 \leq j \leq 2L$, with the convention $P(L+1) = P(L)$,

$$(2.2) \quad A(j) = P(A(j-1)) + P(j - A(j-1)),$$

$$(2.3) \quad B(j) = P(B(j-1) + 1) + P(j - 1 - B(j-1)).$$

All arguments of P lie in $[0, L]$, except that $B(j-1) + 1$ may equal $L+1$.

(4) $A(j) - A(j-1)$ and $B(j) - B(j-1)$ lie in $\{0, 1\}$. Hence $F(u)$ and $G(u)$ are Dyck words of length $2L$.

(5) If $A(j) = a$, $d_j(a) = 1$ and $a-1 \in I_j$, then $u_a = 0$.

Proof. (1) For $a, a+1 \in I_j$,

$$(2.4) \quad d_j(a+1) - d_j(a) = 1 - u_{a+1} + u_{j-a} \in \{0, 1, 2\},$$

so d_j is non-decreasing. At the left end of I_j , $d_j(0) = -P(j) \leq 0$ if $j \leq L$, and $d_j(j-L) = (j-L) - P(j-L) - m \leq \frac{j-L}{2} - m \leq 0$ if $j > L$, using $P(i) \geq i/2$. At the right end, $d_j(j) = j - P(j) \geq 0$ if $j \leq L$, and $d_j(L) = m - P(j-L) \geq 0$ if $j > L$. So both thresholds exist. If $A(j)$ is the left end of I_j then $d_j(A(j)) = 0$; otherwise its predecessor has a negative value and (2.4) gives $d_j(A(j)) \leq 1$. The claim for B is symmetric.

(2) If $a < j/2$ then $d_j(a) \leq a - \frac{a}{2} - \frac{j-a}{2} = a - \frac{j}{2} < 0$, so $A(j) \geq j/2$. For $a = \lceil j/2 \rceil \in I_j$ the same estimate gives $d_j(a) \leq \frac{1}{2}$, hence $d_j(a) \leq 0$ and $B(j) \geq \lceil j/2 \rceil$. Finally $I_{2L} = \{L\}$.

(3) Let $a = A(j-1)$ and $e = d_{j-1}(a)$. By (2), $a \geq \frac{j-1}{2} > j-1-L$, so $a \geq j-L$ and $a \in I_j$; also $1 \leq j-a \leq L$. Directly from the definition,

$$(2.5) \quad d_j(x) = d_{j-1}(x) - u_{j-x}, \quad d_j(x+1) = d_{j-1}(x) + 1 - u_{x+1},$$

so $d_j(x) < 0$ for $x < a$ in I_j , and $d_j(a+1) \geq 0$ when $a+1 \in I_j$. Hence $A(j) \in \{a, a+1\}$. The right side of (2.2) equals $P(a) + P(j-1-a) + u_{j-a} = a - e + u_{j-a}$.

If $e = 0$ and $u_{j-a} = 0$, then $d_j(a) = 0$ and $A(j) = a$. If $e = 0$ and $u_{j-a} = 1$, then $d_j(a) = -1$, so $A(j) = a+1$ (here $a < L$, since $a = L$ would force $P(j-1-L) = m$ and then $P(j-L) = m+1$). If $e = 1$, then a is not the left end of I_{j-1} , and $d_{j-1}(a-1) < 0$ together with (2.4) forces $u_a = 0$ and $u_{j-a} = 1$; then $d_j(a) = 0$ and $A(j) = a$. In every case $A(j) = a - e + u_{j-a}$.

For (2.3), let $b = B(j-1)$ and $e = d_{j-1}(b) \in \{-1, 0\}$. As before $b \geq j-L$ and $0 \leq j-1-b \leq L$. If $e = -1$, then b is not the right end of I_{j-1} , so $d_{j-1}(b+1) > 0$ and (2.4) forces $u_{b+1} = 0$; then $d_j(b+1) = d_{j-1}(b) + 1 = 0$ by (2.5), while $d_j(x) \geq d_{j-1}(x-1) > 0$ for $x > b+1$, so $B(j) = b+1 = P(b+1) + P(j-1-b)$. If $e = 0$ and $b < L$ with $u_{b+1} = 1$, the same computation gives $d_j(b+1) = 0$ and $B(j) = b+1$. If $e = 0$ and either $b = L$ or $u_{b+1} = 0$, then $B(j) = b$, because $d_j(b) \leq d_{j-1}(b) = 0$ and $d_j(b+1) = 1$ when $b < L$. In each case $B(j) = b - e + u_{b+1}$ with $u_{L+1} := 0$, which is (2.3).

(4) follows from (3) and its proof, and with (2) it shows that A and B are prefix counts of Dyck words.

(5) We have $d_j(a-1) < 0$, so by (2.4) $1 - u_a + u_{j-a+1} \geq 2$, giving $u_a = 0$. $\square$

Lemma 2.3. Let u, v be Dyck words of length L .

- (1) Monotonicity. If $u \leq v$ then $F(u) \leq F(v)$ and $G(u) \leq G(v)$.
- (2) If u is irreducible, so is $F(u)$.
- (3) If u is symmetric, so are $F(u)$ and $G(u)$.

Proof. (1) If $P_u \leq P_v$ then $d_j^u(a) \geq d_j^v(a)$ for every a , so both thresholds for u are at most those for v .

(2) Suppose $2A(j) = j$ with $0 < j < 2L$, and put $a = j/2$, so $0 < a < L$. Then $0 \leq d_j(a) = a - 2P(a)$, contradicting irreducibility of u .

(3) Let $a' = L - j + a$. The map $a \mapsto a'$ is an increasing bijection from I_j to I_{2L-j} , and by (2.1),

$$d_{2L-j}(a') = a' - P(L - (j - a)) - P(L - a) = a - P(a) - P(j - a) = d_j(a).$$

Hence $A(2L - j) = L - j + A(j)$ and $B(2L - j) = L - j + B(j)$, which is (2.1) for the outputs, whose half-length is L . $\square$

2.3. The central perturbation. For a symmetric irreducible Dyck word u of length $L = 2m \geq 4$ define

$$T(u) = u_2 u_3 \cdots u_m b (1 - b) u_{m+1} \cdots u_{L-1}, \quad b = 1 - u_m.$$

The first 1 and the last 0 of u are removed and a complementary pair is inserted at the centre. When the centre of u is a peak ($u_m u_{m+1} = 10$) it becomes a valley, and vice versa. Away from the centre, the path of $T(u)$ is the path of u lowered by one level.

Lemma 2.4. *Let u be a symmetric irreducible Dyck word of length $L = 2m \geq 4$, and $u' = T(u)$. Then u' is a symmetric Dyck word and $u' \leq u$. More precisely, $\delta(i) = P_u(i) - P_{u'}(i)$ is given by*

$$(2.6) \quad \delta(i) = \begin{cases} 1 - u_{i+1}, & 0 \leq i < m, \\ u_m, & i = m, \\ u_i, & m < i \leq L. \end{cases}$$

Proof. Formula (2.6) is read off from the definition, using $u_1 = 1$. Since $\delta \geq 0$, $u' \leq u$. The two central letters are complementary and the others are matched by the symmetry of u , so u' is symmetric. The height of u' is $h(i+1) - 1$ for $i < m$ and $h(i-1) - 1$ for $i > m$, where h is the height of u ; these are non-negative because $h > 0$ at interior points and $h(L) = 0$. At the centre the height is $h(m) - 2u_m$, which is non-negative because $h(m) = h(m-1) + 1 \geq 2$ when $u_m = 1$. $\square$

3. THE SYMMETRIC TWO-STEP LEMMA

Lemma 3.1 (Key lemma). *For every symmetric irreducible Dyck word u of length $L = 2m \geq 4$,*

$$G(F(T(u))) \leq T(F(F(u))).$$

The right side is defined because $F(F(u))$ is symmetric and irreducible by Lemma 2.3.

Proof. Let $u' = T(u)$ with prefix counts P' and $\delta = P - P'$ as in (2.6). Let H and K be the prefix counts of $F(u)$ and $F(u')$; by Lemma 2.4 and monotonicity, $K \leq H$. Let W and Z be the prefix counts of $w = F(F(u))$ and $G(F(u'))$. These are symmetric words of length $4L$; put $M = 2L$, the midpoint. For $j \leq M$ the index set in the definition of $W(j)$ and $Z(j)$ is $[0, j]$.

Step 1: a midpoint identity. Let $a = W(M)$. By Lemma 2.1 for $F(u)$, $H(M-a) = L-a+H(a)$, so the threshold value at a is $a - H(a) - H(M-a) = 2a - L - 2H(a)$. It is even and lies in $\{0, 1\}$, so it is 0. Hence $H(a) = a - m$ and

$$H(M - W(M)) = m.$$

Since $j - W(j)$ is non-decreasing in j and H is non-decreasing,

$$(3.1) \quad H(j - W(j)) \leq m \quad (0 \leq j \leq M).$$

Step 2: the contact claim. We show that if $0 \leq j \leq M$ and $Z(j) = W(j)$, then $w_{j+1} = 1$.

Put $a = W(j) = Z(j)$ and $q = j - a$. By Lemma 2.2(1) for W and for Z , and $K \leq H$,

$$0 \leq a - H(a) - H(q) \leq a - K(a) - K(q) \leq 0.$$

So equality holds throughout, and with $K \leq H$ this gives $H(a) = K(a)$ and $H(q) = K(q) =: y$. Put $z = q - y$. By Lemma 2.2(2) applied to H , $y \geq q/2$, so $0 \leq z \leq y$, and $q \leq j/2 \leq L$. By (3.1), $y \leq m$.

By (2.2) for the outer operator, $W(j+1) = H(a) + H(q+1)$, so $w_{j+1} = H(q+1) - H(q)$, the letter of $F(u)$ at position $q+1$. By (2.2) for the inner operator, $H(q+1) = P(y) + P(z+1)$. Writing

$$e = y - P(y) - P(z), \quad e' = y - P'(y) - P'(z),$$

these are the threshold values of $y = H(q)$ for u and of $y = K(q)$ for u' , so $e, e' \in \{0, 1\}$ and

$$(3.2) \quad e' = e + \delta(y) + \delta(z), \quad w_{j+1} = u_{z+1} - e.$$

Suppose $w_{j+1} = 0$, so $u_{z+1} = e$. If $y = 0$ then $q = 0$ and $w_{j+1} = H(1) = 1$, a contradiction; so $y \geq 1$. If $z = m$ then $y = m$ and $q = L$, and $H(L) = m$ says that $F(u)$ has height 0 at the interior point L , contradicting Lemma 2.3(2); so $z < m$. By (2.6), $\delta(z) = 1 - u_{z+1} = 1 - e$, and (3.2) gives $e' = 1 + \delta(y)$. As $e' \leq 1$, we get $e' = 1$ and $\delta(y) = 0$.

Now $\delta(y) = 0$ says $u'_y = 1$: for $y < m$ it gives $u_{y+1} = 1$ and $u'_y = u_{y+1}$, and for $y = m$ it gives $u_m = 0$, so $u'_m = 1 - u_m = 1$. On the other hand, $y = K(q)$ is the first threshold for u' at index q , its value is $e' = 1$, and $y - 1 \in [0, q] = I_q$. Lemma 2.2(5) gives $u'_y = 0$. This contradiction proves the claim.

Step 3: $Z \leq W$. Both start at 0 and have increments 0 or 1. If $Z(j) > W(j)$ for some $j \leq M$, take the least such j ; then $Z(j-1) = W(j-1)$ and $w_j = 0$, contradicting Step 2. So $Z \leq W$ on $[0, M]$, and by Lemma 2.1 applied to both words, $Z(4L-j) - W(4L-j) = Z(j) - W(j)$, so $Z \leq W$ everywhere.

Step 4: domination by $T(w)$. By (2.6) applied to w , the prefix count of $T(w)$ is $W(i) - 1 + w_{i+1}$ for $i < M$ and $W(M) - w_M$ at $i = M$. For $i < M$: if $w_{i+1} = 1$ this is $W(i) \geq Z(i)$; if $w_{i+1} = 0$, Step 2 gives $Z(i) \neq W(i)$, so $Z(i) \leq W(i) - 1$. At $i = M$: if $w_M = 0$ the bound is $Z(M) \leq W(M)$; if $w_M = 1$ then $w_{M+1} = 0$ by symmetry, and Step 2 again gives $Z(M) \leq W(M) - 1$. For $i > M$ the bound follows from the case $4L - i < M$, since $T(w)$ and $G(F(u'))$ are both symmetric. $\square$

Remark 3.2. Symmetry enters through the midpoint identity, which gives the bound $y \leq m$ in Step 2. Without symmetry both fail. Among irreducible Dyck words u , the identity $H(M - W(M)) = m$ first fails at length 12 (for 3 words), and the bound (3.1) first fails at length 16 (for 8 words).

4. THE BLOCK WORDS OF c AND s

For $k \geq 1$ and $L = 2^k$ let w_k be the word of increments $c(L+t) - c(L+t-1)$, $1 \leq t \leq L$. Let v_k be the word of increments of s on the same range when k is even, and the complement of that word when k is odd.

Proposition 4.1. *The sequence s is well defined for all n , and for every $k \geq 1$ the words w_k, v_k are Dyck words with*

$$w_1 = v_1 = 10, \quad w_{k+1} = F(w_k), \quad v_{k+1} = \begin{cases} F(v_k), & k \text{ odd}, \\ G(v_k), & k \text{ even}. \end{cases}$$

Moreover, for $L = 2^k$ and $0 \leq t \leq L$,

$$(4.1) \quad c(L+t) = \frac{L}{2} + P_{w_k}(t), \quad s(L+t) = \begin{cases} \frac{L}{2} + P_{v_k}(t), & k \text{ even}, \\ \frac{L}{2} + t - P_{v_k}(t), & k \text{ odd}. \end{cases}$$

Proof. Define words $\bar{w}_k, \bar{v}_k$ by the recursions in the statement, and define $\bar{c}, \bar{s}$ on $[2, \infty)$ by (4.1) with these words. Since each word is Dyck, the two formulas at $n = 2L$ (the end of block k and the start of block $k+1$) both give L , so $\bar{c}, \bar{s}$ are well defined. We show that $\bar{c}, \bar{s}$, extended by $\bar{c}(1) = \bar{s}(1) = 1$, satisfy the defining recurrences, with every argument on the right in $[1, n-1]$. Strong induction on n then gives $\bar{c} = c$ and $\bar{s} = s$, and shows that s is well defined. The values 1, 2, 2 of c and 1, 1, 2 of s on $[2, 4]$ agree with block 1.

Fix $k \geq 1$, $L = 2^k$, and $n = 2L + j$ with $1 \leq j \leq 2L$. Let u be the parent word ($\bar{w}_k$ or $\bar{v}_k$) with prefix count P and thresholds A, B .

Conway, or s with k odd. Here $\bar{c}(2L+j) = L + A(j)$, and for odd k also $\bar{s}(2L+j) = L + A(j)$, each with A computed from its own parent word. Let $r = A(j-1)$; then $r \in [0, L]$ and

$j - r \in [1, L]$ by the proof of Lemma 2.2(3), so both $L + r$ and $L + j - r$ lie in block k and are less than n . For c ,

$$\bar{c}(\bar{c}(n-1)) + \bar{c}(n - \bar{c}(n-1)) = \bar{c}(L+r) + \bar{c}(L+j-r) = L + P(r) + P(j-r) = L + A(j)$$

by (2.2). For s with k odd, the parent block is given by the second case of (4.1), and

$$n - \bar{s}(L+r) - \bar{s}(L+j-r) = 2L + j - (L+j-P(r) - P(j-r)) = L + A(j).$$

s with k even. Here $\bar{s}(2L+j) = L+j-B(j)$ and the parent is $\bar{s}(L+t) = \frac{L}{2} + P(t)$. Let $r = B(j-1)$, so $\bar{s}(n-1) = L+\rho$ with $\rho = j-1-r \in [0, L]$, and $n - \bar{s}(n-1) = L+r+1$. If $r < L$ this lies in block k and $\bar{s}(L+r+1) = \frac{L}{2} + P(r+1)$. If $r = L$, then $j \geq 2$ and $L+r+1 = 2L+1$ is an earlier point of block $k+1$, where $\bar{s}(2L+1) = L+1-B(1) = L = \frac{L}{2} + P(L+1)$, because $B(1) = 1$ for a Dyck word. In both cases, by (2.3),

$$n - \bar{s}(L+\rho) - \bar{s}(L+r+1) = 2L + j - L - P(j-1-r) - P(r+1) = L + j - B(j).$$

All arguments are at least 1, and $L+r+1 < n$ because $r = L$ forces $j \geq 2$.

The induction runs over j within each block and over k . The words $\bar{w}_k, \bar{v}_k$ are Dyck by Lemma 2.2(4), so they are the increment words described before the proposition. $\square$

Corollary 4.2. *For $L = 2^k$ and $0 \leq t \leq L$,*

$$2c(L+t) - (L+t) = 2P_{w_k}(t) - t, \quad (-1)^k(2s(L+t) - (L+t)) = 2P_{v_k}(t) - t \geq 0.$$

In particular $s(2^k) = c(2^k) = 2^{k-1}$, the sequence s is slow, and $s(n) - \frac{n}{2}$ has sign $(-1)^k$ or vanishes on $[2^k, 2^{k+1}]$. Theorem 1 for $n \geq 2$ is equivalent to $v_k \leq w_k$ for all $k \geq 1$.

Proof. Substitute (4.1). The heights of the Dyck words w_k, v_k are non-negative. $\square$

Lemma 4.3. *Every w_k is symmetric and irreducible, and every v_k is symmetric.*

Proof. The word 10 is symmetric and irreducible. Apply Lemma 2.3(2),(3). $\square$

5. PROOF OF THEOREM 1

The first blocks are computed directly from (2.2) and (2.3):

$$w_2 = v_2 = 1100, \quad w_3 = F(1100) = 11101000, \quad v_3 = G(1100) = 11010100.$$

Deleting the first 1 and the last 0 of w_3 leaves 110100, and inserting 10 at its centre gives $T(w_3) = 11010100 = v_3$.

We prove $v_k \leq T(w_k)$ for every odd $k \geq 3$ by induction. The case $k = 3$ is the equality above. If it holds for an odd k , then by monotonicity (Lemma 2.3(1)) and the key lemma (Lemma 3.1, applicable by Lemma 4.3),

$$v_{k+2} = G(F(v_k)) \leq G(F(T(w_k))) \leq T(F(F(w_k))) = T(w_{k+2}).$$

For the even blocks between them, $v_{k+1} = F(v_k) \leq F(T(w_k)) \leq F(w_k) = w_{k+1}$, using $T(w_k) \leq w_k$ (Lemma 2.4). Together with $v_1 = w_1$ and $v_2 = w_2$, this gives $v_k \leq w_k$ for all $k \geq 1$, and Theorem 1 for $n \geq 2$ follows from Corollary 4.2. For $n = 1$ both sides equal $\frac{1}{2}$. $\square$

6. REMARKS

- (1) The proof gives more than Theorem 1: for odd $k \geq 3$ the path of $|2s(n) - n|$ on $[2^k, 2^{k+1}]$ lies below the path of $2c(n) - n$ lowered by one level away from the ends, with the centre adjusted as in T (Figure 1).
- (2) The argument uses no enumeration. As an independent check, the conclusions of Lemmas 2.2, 2.3, 2.4 and 3.1 were tested by computer on all 26,364 symmetric irreducible Dyck words of length at most 34, Proposition 4.1 on every block up to 2^{20} , and Theorem 1 and Corollary 4.2 on all $n \leq 2^{20}$.

- (3) Every result of Sections 2 to 5 is also proved in the Lean 4 proof assistant with the Mathlib library: Lemma 2.1, Lemmas 2.2 to 3.1, Proposition 4.1 (including that c and s satisfy their recurrences with every argument in range), Corollary 4.2, Lemma 4.3, Theorem 1, and the sharper statement of Remark 1. The proofs follow the paper; the base case $T(w_3) = v_3$ is checked by evaluation, as in Section 5. They use no axioms beyond the standard ones, and sixteen deliberately altered statements and definitions are rejected by the proof checker. The Lean code and the verification program are at <https://github.com/jvvk/mathematics/tree/main/hofstadter-conway-comparison>.
- (4) Alkan also asked for a combinatorial description of s . Proposition 4.1 gives one in the same spirit as the analysis of c by Kubo and Vakil: each block of s is obtained from the previous one by one of two explicit threshold operators, alternating with the parity of k .

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